Abdelrahman Khalifa

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Summary

Software engineer who programs for fun as much as for work: 11 PRs merged upstream as a Google Summer of Code 2026 contributor with The Linux Foundation (OpenPrinting), plus merged changes in Zephyr RTOS, Rust Clippy and the Boa JavaScript engine. Writes Python, Rust, C and C++; uses Linux every day. Teaching himself database internals through Carnegie Mellon's database course. Codeforces Expert.

Experience

Google Summer of Code Contributor May 2026 – Oct 2026
The Linux Foundation (OpenPrinting) *Remote*

- Built the **Printers page for COSMIC Settings** and a standalone app with the same UI (cosmic-settings #2182, in review).
- **11 PRs merged upstream** (list): ported cups-rs to libcups3 with a CI matrix across architectures and CUPS versions, added DNS-SD bindings, and fixed crashes and a leaked client in libcups (C).

Embedded Software Engineer Aug 2025 – Present
EKSON *Egypt*

- Developed the Python (PyQt5) control app; used asyncio to wait for ESP32 and server messages concurrently, cutting CPU use and latency by about 40% (measured).
- Built remote operation and firmware updates: a Node.js/Express + MongoDB backend that the desktop app polls over REST to relay commands and flash new firmware; GitHub Actions CI.

Embedded Linux Systems Intern Dec 2024 – Jan 2025
Orange Digital Center (ODC) *Egypt*

- Built custom Linux images with Yocto and Buildroot and wrote a Linux kernel module; scored 100% in the final evaluation.

OSS Projects

Zephyr RTOS, Boa JavaScript engine (Rust), Rust Clippy | *Contributor (all merged)* Mar 2025 – Dec 2025

- Zephyr RTOS: added the `IP_MULTICAST_LOOP` socket option for IPv4 in 6 networking modules (#86977).
- Boa: optimized `JsStr::code_points()` for Latin1 strings, avoiding UTF-16 conversion (PR #4553).
- Rust Clippy: fixed `ref_as_ptr` false positives for non-temporary references in `let/const` (PR #16214).

Projects

Database Systems, CMU 15-445 | *C++17, SQL* GitHub

- Thread-safe Count-Min Sketch for the BusTub database (per-row locks, merge, top-k); SQL analytics in DuckDB (joins, aggregation, upserts).

Function Solver and Plotter | *Python, PySide2, SymPy, Matplotlib* GitHub

- Solves and plots two functions with their intersections; input validation and end-to-end tests with pytest-qt.

Technical Skills

Languages: Python (asyncio), Rust, C, Modern C++, SQL, JavaScript (Node.js), Bash
Data & backend: SQL (DuckDB, SQL Server), MongoDB, REST APIs, Express; database internals (storage, buffer pool, hash tables)
Linux & tools: Linux daily driver (Arch-based CachyOS), kernel modules, Git/GitHub, GitHub Actions CI, Cargo, CMake, pytest, debugging, code review

Education

Minia University Minia, Egypt
Bachelor of Science (B.Sc.), Computer and Systems Engineering *2020 – 2025*

- Grade: Very Good with Honors.
- Codeforces Expert (handle _Khalifa); 4-time ECPC finalist, 2021–2024; mentor in the local ICPC community.